

Packed Weeks, Not Back-to-Backs: Schedule Density and Time-to-Absence in NBA Rotation Players

Introduction

The NBA has cut back-to-backs and tied awards to games played, yet evidence linking schedule congestion to player availability is mixed: back-to-backs alone have not predicted game injuries, while the condensed 2020 season showed higher injury incidence. Most studies aggregate exposure at the season or team level. We model schedule density as a game-level, time-varying exposure and ask whether games in a rolling 7-day window change the hazard of losing a rotation player.

Methods

We used public ESPN box scores (sportsdataverse) for seven regular seasons (2018-19 to 2025-26, excluding 2020-21, the Omicron wave, the 2020 bubble and each team's last 10 games). Rotation players (5+ prior games, 15+ prior minutes per game) contributed 103,878 player-games. Because public box scores omit inactive players, the outcome was an injury-proxy absence: a player appears in a game and then misses his team's next three or more games, censoring at trades and season end. Andersen-Gill recurrent-event Cox models on team-game time, stratified by season with player-clustered errors, included games in the past 7 days, rest (back-to-back, 1 day, 2+ days), home/away and player role (prior average minutes). Same-game minutes were not adjusted because in-game injuries truncate them. Secondary models added travel (km in 7 days, time-zone shift). Sensitivity analyses used a player frailty, restored excluded games, and restricted events to those confirmed by an injury/illness reason in the box score. All code and data are public.

Results

We observed 1,748 absences among 808 players (16.8 per 1,000 player-games; median 5 games missed, IQR 4-10). The proxy behaved like an injury marker: 23.0% of games preceding an absence ended below half the player's usual minutes, versus 3.7% of other games (Figure 2). Each additional game in the 7-day window raised the hazard by 11% (HR 1.11, 95% CI 1.01-1.22); a 4th game in 7 days versus a 3rd gave HR 1.10 (0.98-1.24) (Figure 1). The second night of a back-to-back showed no excess risk (HR 0.94, 0.82-1.08), and travel distance was null (HR 1.00 per 1,000 km) without changing the density estimate. Heavier-role players were at higher risk (HR 1.17 per 10 prior minutes). More minutes in the previous six days predicted lower hazard (HR 0.86 per 30 minutes, 0.81-0.92), consistent with healthy-player selection: teams play players who are fit. The density estimate held with frailty, attenuated when excluded games were restored (4+ vs 3: HR 1.06, 0.95-1.18), and was imprecise among the 476 confirmed injuries (HR 1.06 per game, 0.89-1.26).

Conclusion

In public data, cumulative weekly density, not the back-to-back or travel, carried a modest, fragile availability signal. Recent minutes predicted lower risk, likely because rest decisions select who is exposed. For scheduling, spreading games within weeks may matter more than removing individual back-to-backs; for performance staff, recent-minutes metrics confound readiness with risk. The full paper will validate outcomes against official injury reports and emulate a target trial of rest on next-week hazard.

Schedule density and hazard of an injury-proxy absence

NBA 2018-19 to 2025-26 (excl. 2020-21), rotation players, Cox model stratified by season

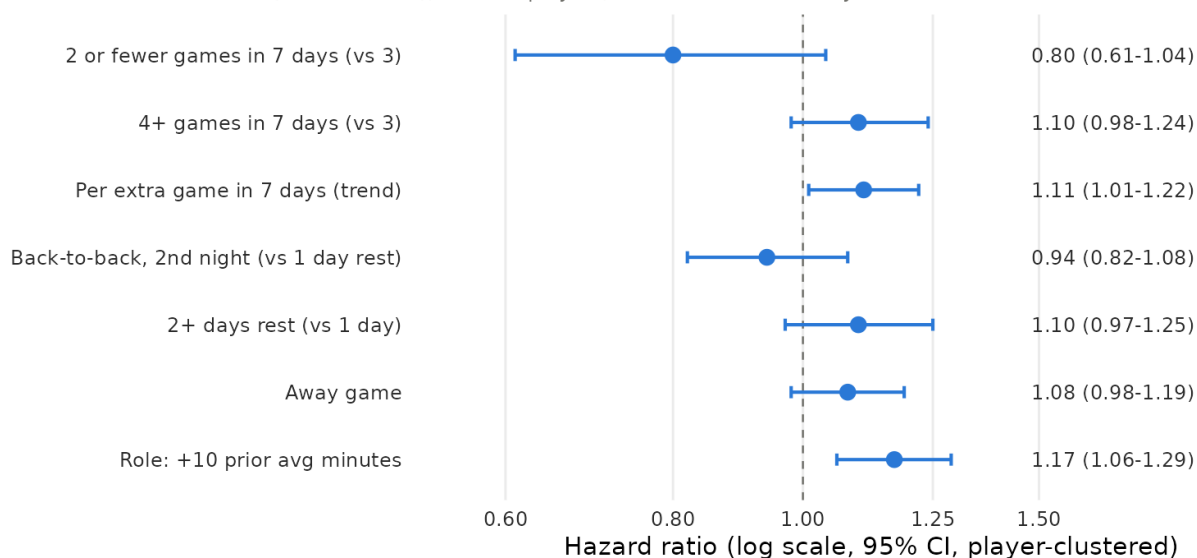


Figure 1. Adjusted hazard ratios for an injury-proxy absence (Andersen-Gill Cox model, stratified by season, player-clustered 95% CI).

Absences are preceded by truncated games

23% of pre-absence games fall below half the player's usual minutes vs 3.7% of other games

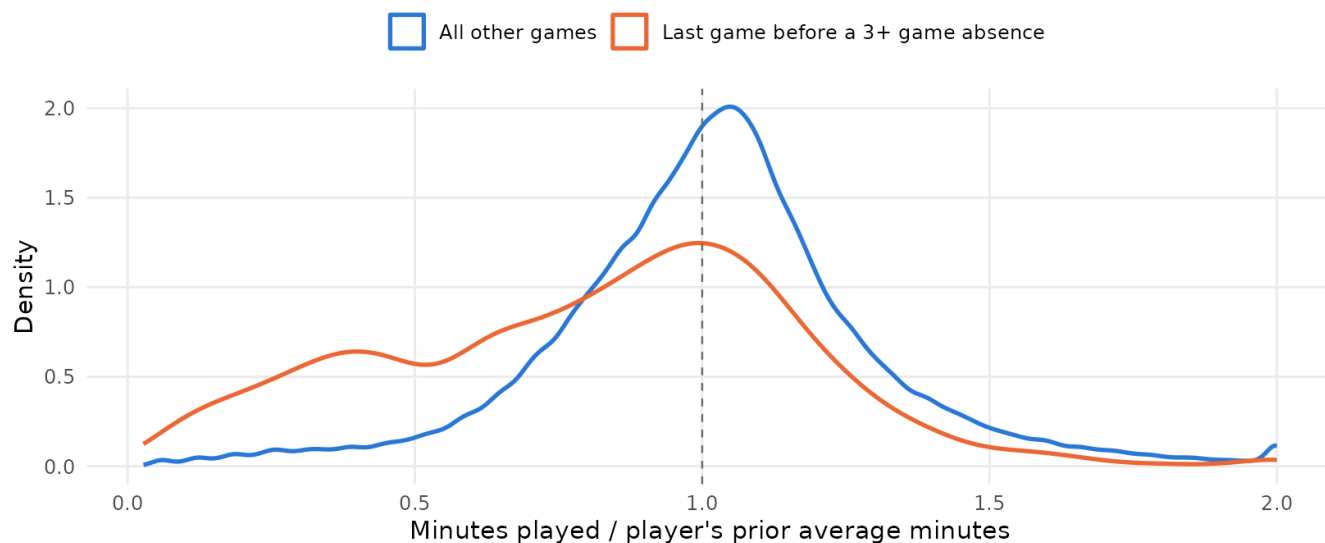


Figure 2. Face validity of the absence proxy: minutes played relative to the player's prior average in the last game before a 3+ game absence vs all other games.

Code and data: github.com/AdrianRubioJ/nba-schedule-density-absence